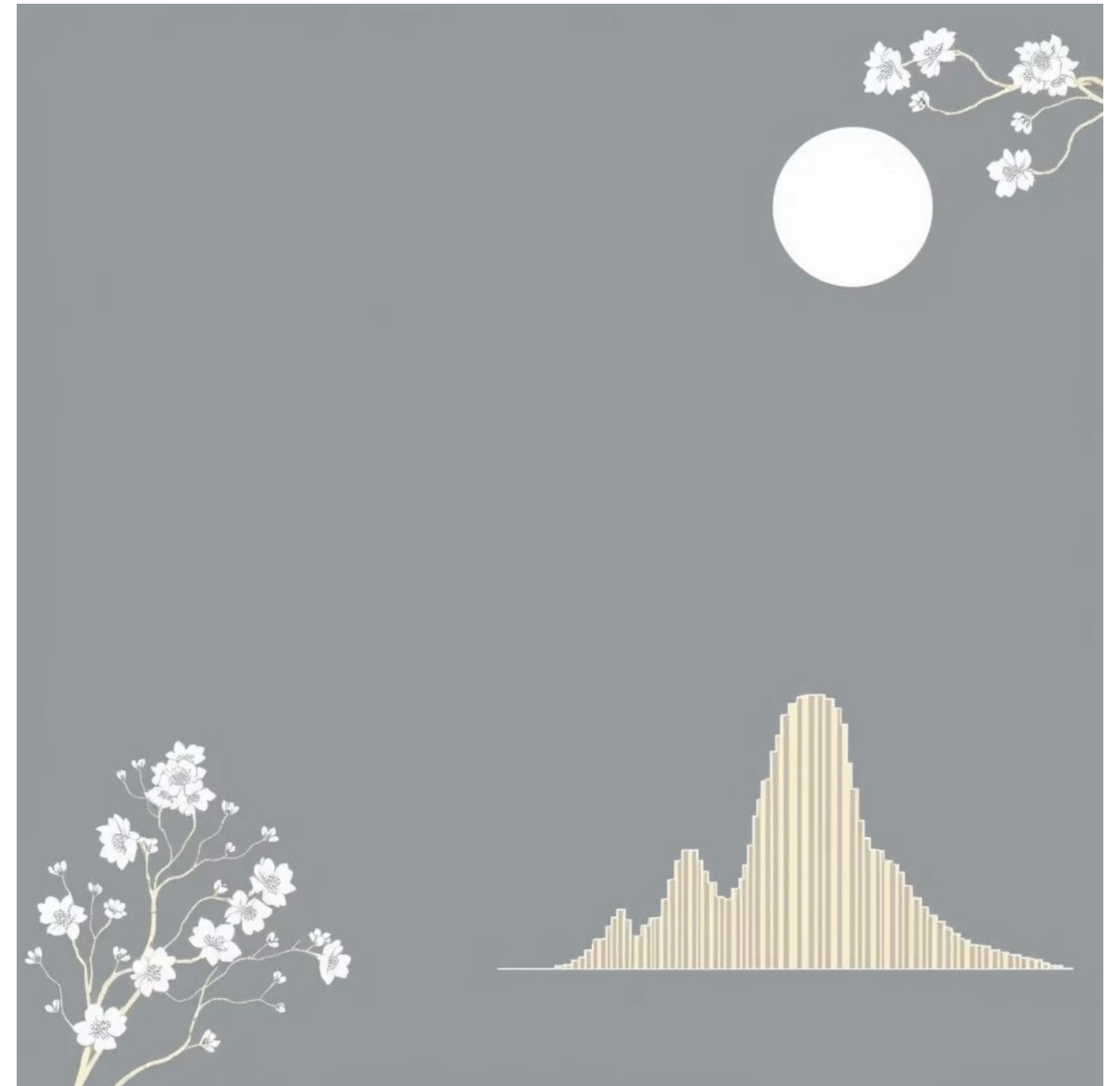
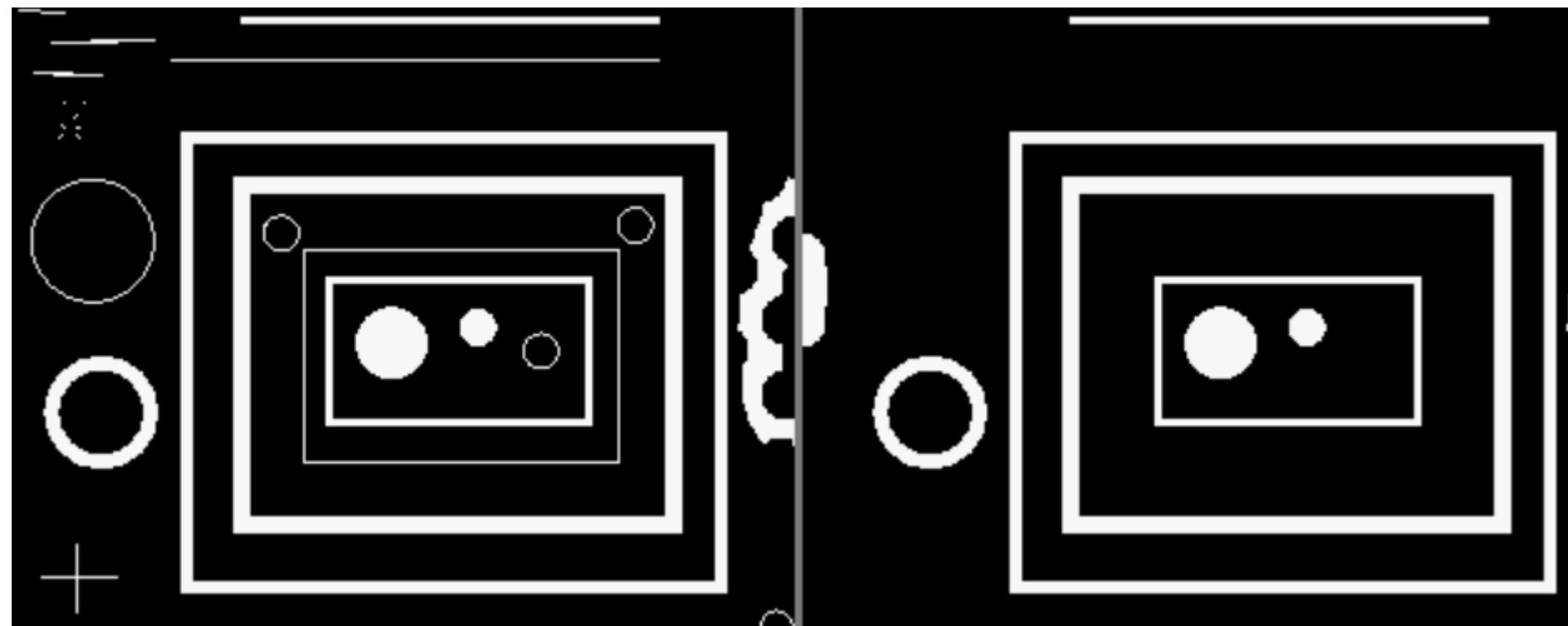
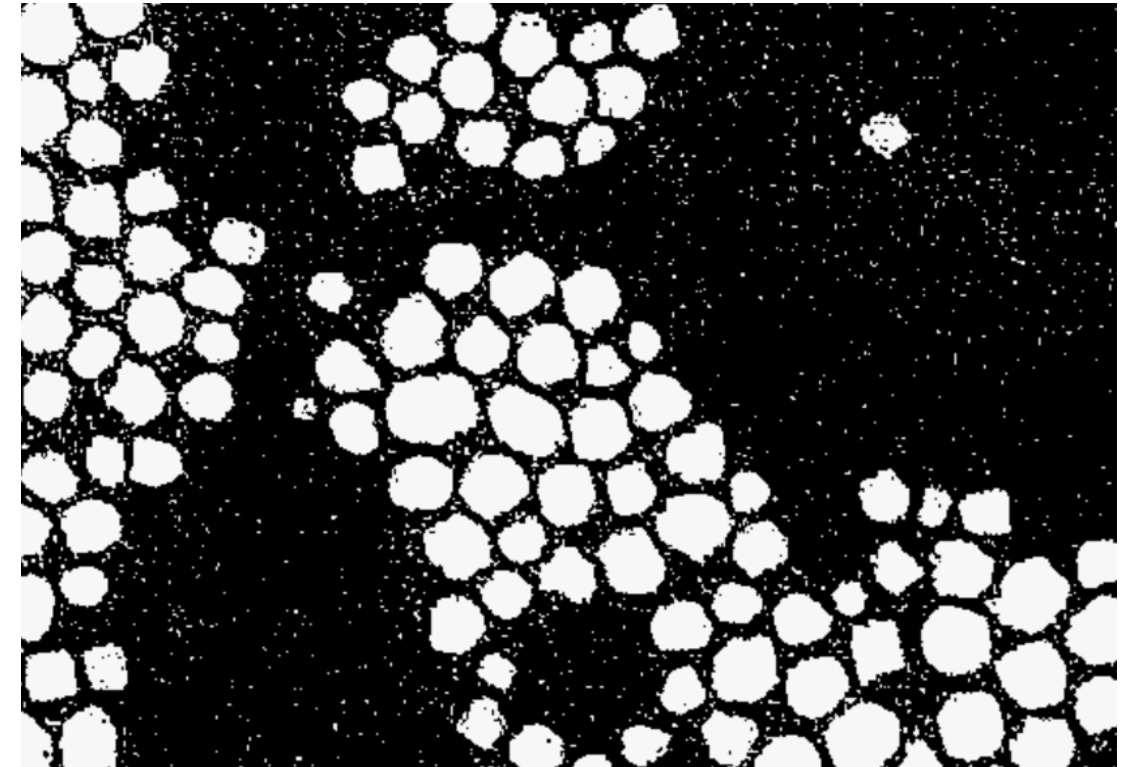


Morphological image processing ?

- Morphological operations are image-processing techniques used to analyze and process geometric structures in binary and grayscale images.
- These operations focus on the shape and structure of objects within an image. They are particularly useful in image segmentation, object detection, and noise removal tasks.
- Morphological operations aim to probe and transform an image based on its shape, enhancing features or removing imperfections.





Description: Morphological Image Processing

Definition: Morphological processing deals with the shape and structure of objects in binary or grayscale images.

Historical Background: Developed in the late 20th century, influenced by set theory and topology.

Essential Concepts:

Structuring Element: A small shape or template used to probe the input image. It determines the nature of the operation and is typically a small binary matrix.

Description: Morphological Image Processing

1	1	1	1	1	1	1	
			1	1	1	1	
			1	1	1	1	
		1	1	1	1	1	
			1	1	1	1	
		1	1				

Structuring Element-S

1	1	1
1	1	1
1	1	1

[illegible]

Fundamental Operations

Morphological image processing mainly involves a set of basic operations:

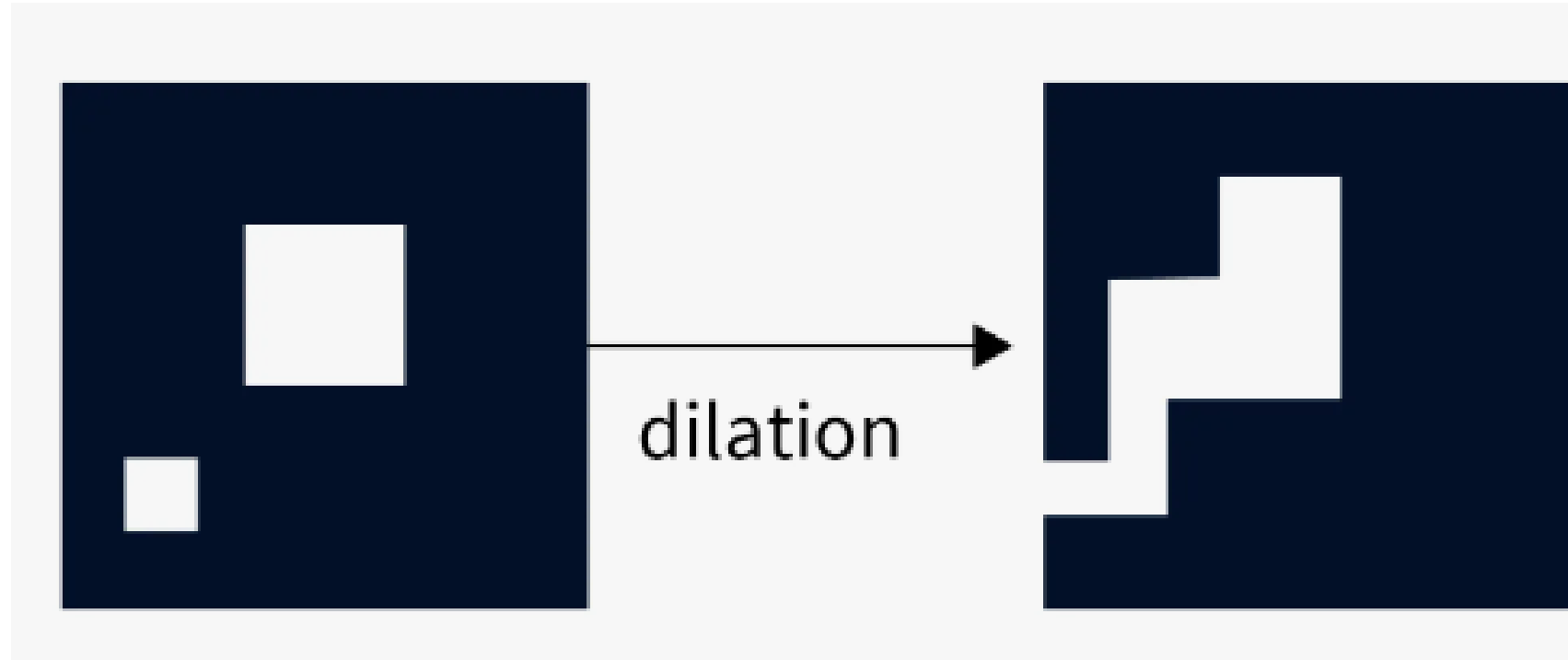
Dilation

Definition: Dilation adds pixels to the boundaries of objects in an image.

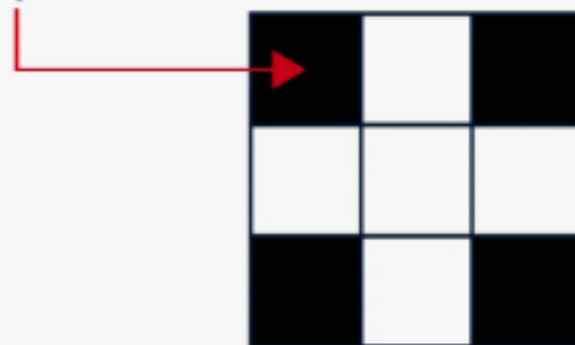
Operation: For each pixel in the image, if any pixel under the structuring element is on, the corresponding pixel in the output image is set to on.

Effect: Helps to fill small holes and connect disjoint parts of objects.

Dilation process



Anchor point



The structuring used

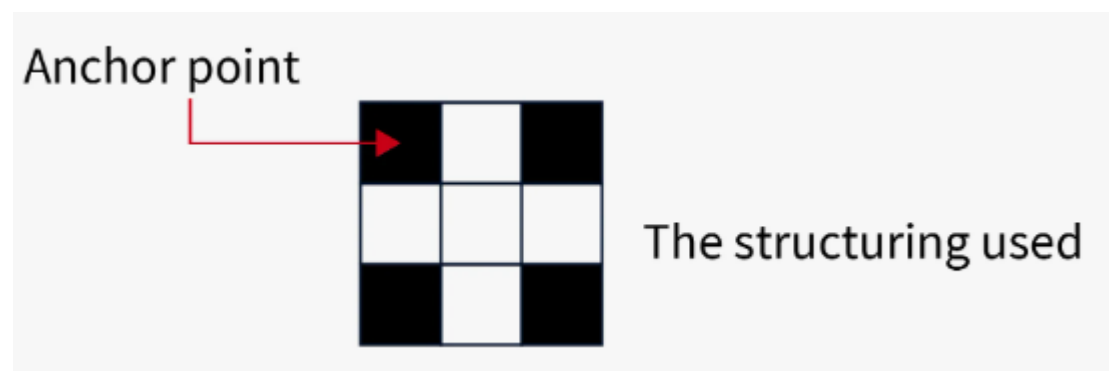
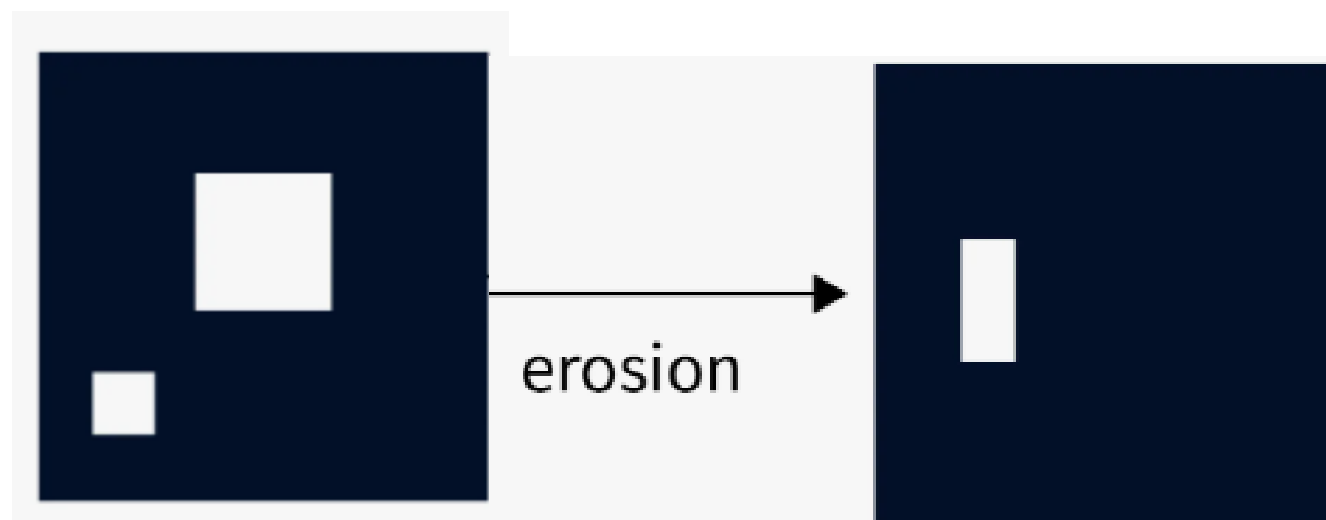
Erosion Process

Definition: Erosion removes pixels on object boundaries.

Operation: The output is a pixel if the structuring element fits completely within the object.

Effect: Useful for removing small-scale noise and separating objects that are close together.

Erosion Process



Opening process

Definition: Opening is an operation that involves erosion followed by dilation.

Effect: Smoothens the contours of an object, removes small objects, and breaks connections between objects.

Use Case: Helpful for noise reduction without affecting the size of larger objects significantly.

Closing

Definition: Closing is dilation followed by erosion.

Effect: Helps in closing small holes and gaps in objects while preserving the overall shape.

Use Case: Useful for filling in small holes in an object.

Summary

- Morphological operations are a powerful, non-linear alternative to convolution.
- They rely on the Structuring Element to probe the image's geometry.
- The four main operations (Dilation, Erosion, Opening, Closing) are essential tools for manipulating object shape, size, and connectivity in digital images, providing crucial pre-processing steps for advanced AI tasks.

